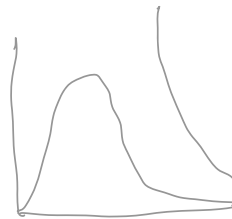


LECTURE 1

FOUNDATIONS OF QUANTUM COMPUTING

github.com/barghouthi/ssft26



Max Planck
plankmaxxing



① Integer factoring Shor 94

② quantum chemistry simulation Lloyd 96

State

0 ~~0~~

1 ~~0~~

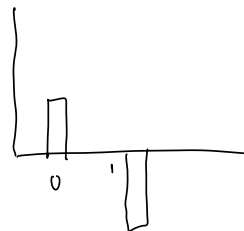
qubit

$$\begin{matrix} 0 \\ 1 \end{matrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

e.g. $\begin{matrix} 0 \\ 1 \end{matrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

amplitude
 $\begin{matrix} 0 \\ 1 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$

$$\left| \frac{1}{\sqrt{2}} \right|^2 + \left| \frac{1}{\sqrt{2}} \right|^2 = 1$$



x	a
00	$1/2$
01	$1/2$
10	$1/2$
11	$1/2$

operations (gates)

① reversible

② normalizing

X gate (NOT)

$$x \rightarrow \neg x$$

$$|x\rangle \rightarrow |\neg x\rangle$$

$$\begin{array}{c} 0 \\ 1 \end{array} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \xrightarrow{\text{X}} \begin{array}{c} 0 \\ 1 \end{array} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{array}{c} 0 \\ 1 \end{array} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \xrightarrow{\text{X}} \begin{array}{c} 0 \\ 1 \end{array} \begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\begin{array}{cc} 00 & \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \end{array} \longrightarrow \begin{array}{cc} 10 & \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \end{array}$$

$$\begin{array}{cc} 01 & \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \end{array} \longrightarrow \begin{array}{cc} 11 & \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \end{array}$$

$$\begin{array}{cc} 10 & \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \end{array} \longrightarrow \begin{array}{cc} 00 & \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \end{array}$$

$$\begin{array}{cc} 11 & \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \end{array} \longrightarrow \begin{array}{cc} 01 & \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \end{array}$$

CX (CNOT) / controlled NOT

control target

$$\begin{array}{cc} 00 & \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \end{array} \longrightarrow \begin{array}{cc} 00 & \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \end{array}$$

$$\begin{array}{cc} 01 & \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \end{array} \longrightarrow \begin{array}{cc} 01 & \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \end{array}$$

$$\begin{array}{cc} 10 & \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \end{array} \longrightarrow \begin{array}{cc} 11 & \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix} \end{array}$$

$$\begin{array}{cc} 11 & \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \end{array} \longrightarrow \begin{array}{cc} 10 & \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \end{array}$$

if control then flip target

updating amplitudes (rotations)

$$\begin{array}{c} 0 \\ 1 \end{array} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \longrightarrow \begin{array}{c} 0 \\ 1 \end{array} \begin{bmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{bmatrix}$$

$$|x\rangle \longrightarrow i^x |x\rangle$$

$$|c| = 1$$

$$|i| = 1$$

Hadamard gate

$$\begin{matrix} 0 \\ 1 \end{matrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \begin{matrix} \rightarrow 0 \\ \rightarrow 1 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{matrix} 0 \\ 1 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$|0\rangle \rightarrow \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$|1\rangle \rightarrow \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

Measurement

what is the amplitude?

Is the qubit 0 or 1?

$$\begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} \xrightarrow{\text{CNOT}} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix} \rightarrow \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \\ 0 \end{bmatrix}$$

Entanglement

$$\begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 0 \\ \cancel{1/\sqrt{2}} \end{bmatrix} \rightarrow \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

EPR / Bell

$$\begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

H_0

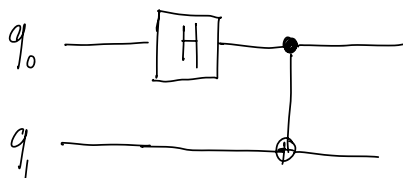
$$\begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

$CX_{0,1}$

$$\begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

$$00 \xrightarrow{1/\sqrt{2}} 00$$

$$\xrightarrow{1/\sqrt{2}} 10$$



Evaluate function $f: \{0,1\} \rightarrow \{0,1\}$

$$f(x) = 1$$

$$f(x, y) = (x, y \oplus 1)$$

input $\uparrow$ output $\uparrow$

usually 0

Parallel Evaluation

$$\begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

f

$$\begin{matrix} 0 & f(0) \\ 0 & \neg f(0) \\ 1 & f(1) \\ 1 & \neg f(1) \end{matrix}$$

$$\begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$$

Deutsch (-Jozsa)

$f: \{0,1\} \rightarrow \{0,1\}$ is it constant/balanced.

$$00 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \xrightarrow{X_1} \xrightarrow{H_0, H_1} \begin{matrix} 00 \\ 01 \\ 10 \\ 11 \end{matrix} \begin{bmatrix} 1/2 \\ -1/2 \\ 1/2 \\ -1/2 \end{bmatrix} \begin{bmatrix} + \\ - \\ + \\ - \end{bmatrix}$$

Phase Kickback

$$\begin{matrix} 0 & f(0) \\ 0 & \neg f(0) \\ 1 & f(1) \\ 1 & \neg f(1) \end{matrix} \begin{bmatrix} + \\ - \\ + \\ - \end{bmatrix}$$

constant

$$\begin{matrix} 0 & f(0) \\ 0 & \neg f(0) \\ 1 & f(1) \\ 1 & \neg f(1) \end{matrix} \begin{bmatrix} + \\ - \\ 0 \\ 0 \end{bmatrix}$$

$$f(x) = 1$$

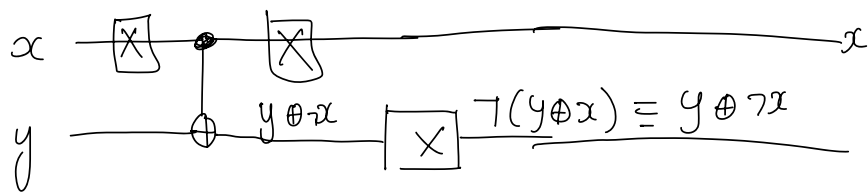
$$f(x, y) = (x, y \oplus 1)$$

$$x \text{ ————— } x$$

$$0 \text{ — } \boxed{\times} \text{ — } 1$$

$$f(x) = \neg x$$

$$f(x, y) = (x, y \oplus \neg x)$$



$$\neg \neg x \equiv x$$

$$\neg(1 \oplus x) \equiv 1 \oplus \neg x$$

$$x \equiv x$$